CIVITAS

Sustainable and smart mobility for all

2030

CIVITAS indicators

Average time per delivery (TRA_FR_EF3)

DOMAIN

Transport	Environment	Energy	Society	Economy

TOPIC	**Freight**	
IMPACT	**Operational efficiency of urban deliveries** *Improved efficiency and reliability of last-mile delivery operations.*	**TRA_FR**

Category

Key indicator	Supplementary indicator	State indicator

CONTEXT AND RELEVANCE

Urban freight measures frequently aim to improve the efficiency and productivity of delivery operations while maintaining service quality. A reduction in average time per delivery indicates more efficient logistics organisation, improved route planning, reduced congestion impacts, better stop management, increased delivery consolidation, or the use of urban freight solutions that reduce operational delays. These improvements can help logistics operators complete deliveries more efficiently while reducing operating costs, congestion impacts, emissions, and pressure on scarce public space.

This indicator measures the average time required to complete a delivery within the pilot area. **It is a relevant indicator when the policy action aims to increase the efficiency of urban freight distribution. A successful action is reflected in a LOWER value of the indicator after the experiment compared to the BAU case.**

DESCRIPTION

The indicator assesses how efficiently delivery operations are performed and whether freight interventions reduce the amount of time required to complete each delivery. The time considered should include the complete operational time associated with delivery tours, including driving time, waiting time, parking search time and loading and unloading activities.

The indicator is expressed as **min/delivery**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **exogenously** based on the required inputs and then coded in the supporting tool**.**

Method 1

Survey delivery companies on a sample of delivery tours	Significance: **0.50**

INPUTS

The following information is needed to compute the indicator:

- The **total number of deliveries** made in the pilot area by the surveyed logistics operators over a defined reference period, including only motorised modes.
- The **total operational time** spent by the motorised freight vehicles that performed the recorded deliveries during the reference period.

EQUATIONS

The equation that should be applied to quantify the indicator is:

$$\boldsymbol{AvgTimeDel} = \frac{\sum_i Time_i}{\sum_i Deliveries_i}$$

Where:

$AvgTimeDel$ = Average time per delivery (min/delivery)

$Time_i$= Operational time spent by delivery vehicle i during the reference period (minutes)

$Deliveries_i$= Number of completed deliveries made by vehicle i during the reference period

ALTERNATIVE INDICATORS

This indicator measures the average time required to complete a delivery. Other indicators in the CIVITAS Evaluation Framework addressing the operational efficiency of urban deliveries include: **TRA_FR_EF1**, which measures the average distance travelled per delivery; **TRA_FR_EF2**, which measures the share of unsuccessful delivery attempts; and **TRA_FR_EF4** and **TRA_FR_EF5**, which measure the average load factor of delivery vehicles for B2B and B2C deliveries, respectively.